JAMA | Original Investigation

Ivonescimab Plus Chemotherapy in Non–Small Cell Lung Cancer With *EGFR* Variant
A Randomized Clinical Trial

HARMONi-A Study Investigators

IMPORTANCE For patients with non–small cell lung cancer whose disease progressed while receiving EGFR tyrosine kinase inhibitor (EGFR-TKI) therapy, particularly third-generation TKIs, optimal treatment options remain limited.

OBJECTIVE To compare the efficacy of ivonescimab plus chemotherapy with chemotherapy alone for patients with relapsed advanced or metastatic non–small cell lung cancer with the epidermal growth factor receptor (*EGFR*) variant.

DESIGN, SETTING, AND PARTICIPANTS Double-blind, placebo-controlled, randomized, phase 3 trial at 55 sites in China enrolled participants from January 2022 to November 2022; a total of 322 eligible patients were enrolled.

INTERVENTIONS Participants received ivonescimab (n = 161) or placebo (n = 161) plus pemetrexed and carboplatin once every 3 weeks for 4 cycles, followed by maintenance therapy of ivonescimab plus pemetrexed or placebo plus pemetrexed.

MAIN OUTCOMES AND MEASURES The primary end point was progression-free survival in the intention-to-treat population assessed by an independent radiographic review committee (IRRC) per Response Evaluation Criteria in Solid Tumors version 1.1. The results of the first planned interim analysis are reported.

RESULTS Among 322 enrolled patients in the ivonescimab and placebo groups, the median age was 59.6 vs 59.4 years and 52.2% vs 50.9% of patients were female. As of March 10, 2023, median follow-up time was 7.89 months. Median progression-free survival was 7.1 (95% CI, 5.9-8.7) months in the ivonescimab group vs 4.8 (95% CI, 4.2-5.6) months for placebo (difference, 2.3 months; hazard ratio [HR], 0.46 [95% CI, 0.34-0.62]; $P < .001$). The prespecified subgroup analysis showed progression-free survival benefit favoring patients receiving ivonescimab over placebo across almost all subgroups, including patients whose disease progressed while receiving third-generation EGFR-TKI therapy (HR, 0.48 [95% CI 0.35-0.66]) and those with brain metastases (HR, 0.40 [95% CI, 0.22-0.73]). The objective response rate was 50.6% (95% CI, 42.6%-58.6%) with ivonescimab and 35.4% (95% CI, 28.0%-43.3%) with placebo (difference, 15.6% [95% CI, 5.3%-26.0%]; $P = .006$). The median overall survival data were not mature; at data cutoff, 69 patients (21.4%) had died. Grade 3 or higher treatment-emergent adverse events occurred in 99 patients (61.5%) in the ivonescimab group vs 79 patients (49.1%) in the placebo group, the most common of which were chemotherapy-related. Grade 3 or higher immune-related adverse events occurred in 10 patients (6.2%) in the ivonescimab group vs 4 (2.5%) in the placebo group. Grade 3 or higher vascular endothelial growth factor–related adverse events occurred in 5 patients (3.1%) in the ivonescimab group vs 4 (2.5%) in the placebo group.

CONCLUSIONS Ivonescimab plus chemotherapy significantly improved progression-free survival with tolerable safety profile in TKI-treated non–small cell lung cancer.

TRIAL REGISTRATION ClinicalTrials.gov Identifier: NCT05184712

Visual Abstract

Supplemental content

Group Information: The HARMONi-A Study Investigators appear at the end of the article.

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The standard first-line therapy for patients with non-small cell lung cancer (NSCLC) with the epidermal growth factor receptor (*EGFR*)-sensitizing variant has been EGFR tyrosine kinase inhibitors (EGFR-TKIs). Notably, third-generation EGFR-TKIs, including osimertinib, have been approved as a first-line treatment for patients with advanced NSCLC with the *EGFR* variant.[1-3] However, acquired resistance to osimertinib, similar to early-generation EGFR-TKIs, inevitably occurs.[4] For patients whose illness has progressed while receiving EGFR-TKIs therapy, especially with third-generation TKIs, optimal treatment options remain limited.

Although immune checkpoint inhibitors combined with chemotherapy are the standard therapy for patients with wild-type NSCLC, anti-programmed death-1 (anti-PD-1) combined with chemotherapy has shown limited efficacy in patients with *EGFR*-variant NSCLC whose illness progressed while receiving EGFR-TKI therapy, both KEYNOTE-789 and CHECKMATE-722 studies failed to show improved efficacy in this population.[5,6] Preclinical studies have demonstrated the synergistic effects of vascular endothelial growth factor (VEGF) inhibitors and immune checkpoint inhibitors,[7-9] and studies evaluating immunotherapy in combination with antiangiogenic agents and chemotherapy have shown promising results. Results of the subgroup analyses from the IMpower150 trial demonstrated that the addition of atezolizumab to bevacizumab and chemotherapy significantly improved progression-free survival (PFS).[10] Similarly, the ATTLAS study further confirmed the efficacy of atezolizumab plus bevacizumab, carboplatin, and paclitaxel in patients whose illness progressed while receiving TKI treatment.[11] The ORIENT-31 trial showed that sintilimab plus bevacizumab biosimilar IBI305, pemetrexed, and cisplatin resulted in significant improvement in PFS compared with chemotherapy alone for patients whose illness progressed while receiving EGFR-TKI therapy.[12] Although, the IMpower151 study did not meet the primary end point, which also used the 4 drug combination regimen (atezolizumab plus bevacizumab, carboplatin, and pemetrexed or paclitaxel), the subgroup analysis showed that patients who experienced resistance to first- or second-generation EGFR-TKI treatment could benefit more, with a hazard ratio (HR) of 0.55 (95% CI, 0.29-1.02).[13] However, the proportion of patients who received the third-generation EGFR-TKIs in these studies was very low.

Ivonescimab (AK112/SMT112) has been developed as a first-in-class, humanized tetravalent bispecific antibody. The in vitro studies have shown that ivonescimab possesses great binding affinity to PD-1 and VEGF.[14] The first human study demonstrated favorable safety and antitumor activity in solid tumors,[15] a phase 2 study in patients with NSCLC demonstrated the promising efficacy of ivonescimab plus chemotherapy in those whose disease progressed while receiving EGFR-TKI therapy with an objective response rate (ORR) of 49.8% and median PFS of 12 months, and the safety profile was shown to be tolerable and manageable.[16]

The HARMONi-A (AK112-301) study is the first double-blind, placebo-controlled, randomized phase 3 study to compare the efficacy of ivonescimab plus chemotherapy with chemotherapy alone for patients with advanced or metastatic *EGFR*-variant NSCLC who showed disease progression after EGFR-TKI treatment. The results of the first preplanned interim analysis are reported in this article.

Key Points

Question Among patients with non–small cell lung cancer with the epidermal growth factor receptor (*EGFR*) variant who experienced resistance to EGFR tyrosine kinase inhibitor (EGFR-TKI) treatment, does the addition of ivonescimab (a bispecific antibody targeting programmed cell death 1 protein and vascular endothelial growth factor) to chemotherapy improve progression-free survival compared with chemotherapy alone?

Findings In this phase 3 trial that enrolled 322 patients whose disease progressed while receiving EGFR-TKI treatment, ivonescimab plus chemotherapy significantly improved progression-free survival compared with placebo plus chemotherapy. Median progression-free survival was 7.06 months with ivonescimab vs 4.80 months with placebo.

Meaning Ivonescimab plus chemotherapy significantly improved progression-free survival with tolerable safety profile in patients with non–small cell lung cancer who previously underwent EGFR-TKI treatment and may offer a new treatment option for patients with TKI resistance.

Methods

Study Design and Participants

This randomized, double-blind, placebo-controlled, phase 3 study was conducted at 55 study sites in China. Eligible participants were adults aged 18 to 75 years with locally advanced or metastatic NSCLC (stage III B, III C, or IV per the *American Joint Committee on Cancer Staging Manual, Eighth Edition*) and *EGFR* variant whose disease had progressed after EGFR-TKI therapy. Disease progression was defined as progression after receiving first-generation or second-generation EGFR-TKIs and having a confirmed *EGFR* Thr790Met negative variant status or after receiving a third-generation EGFR-TKI as first- or second-line treatment. Other inclusion criteria included an Eastern Cooperative Oncology Group performance status score of 0 or 1 with at least 1 measurable lesion (per Response Evaluation Criteria in Solid Tumors [RECIST], version 1.1). Patients were excluded if they had small cell or squamous cell histology, previously received systemic antitumor therapy (including cytotoxic chemotherapy) except EGFR-TKIs, or had previously received immunotherapy (including anti-PD-1, anti-PD-L1, anti-PD-L2, or anti-cytotoxic T-lymphocyte associated protein 4) antibodies or agents that act on T-cell costimulation or other immune checkpoint pathways. The full eligibility criteria were listed in the study protocol (Supplement 1).

The study was conducted in accordance with the International Conference on Harmonization Good Clinical Practice guidelines and the Declaration of Helsinki. The study protocol was approved by the ethics review board at Sun Yat-Sen University Cancer Center and independent ethics committees at

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each study site. All participants provided written informed consent before screening. This study was reported following the CONSORT reporting guideline by meeting requirements from CONSORT 2010 checklist.

Randomization and Masking

Eligible patients were randomly assigned in a 1:1 ratio by stratified blocked randomization using interactive web/voice response system to receive either ivonescimab plus chemotherapy (pemetrexed and carboplatin) or placebo plus chemotherapy. Patients were stratified according to the third-generation EGFR-TKI (received vs not received) and brain metastases (presence vs absence). Because this study was double-blind, patients, investigators, and other study personnel including review committee and sponsor's study team were masked to treatment randomization. The placebo was packaged and administered in the same manner as ivonescimab during the treatment and maintenance periods.

Procedures

Patients received all study drugs intravenously on day 1 and then once every 3 weeks for 4 cycles, including ivonescimab 20 mg/kg or placebo (Akeso Biopharma, Inc), pemetrexed 500 mg/m^2, and carboplatin AUC 5 mg/mL/min, followed by maintenance therapy of ivonescimab plus pemetrexed or placebo plus pemetrexed. Dose modification was only allowed for chemotherapy. Treatment was continued until disease progression, intolerable toxic effects, investigator decision, patient withdrawal of consent, or death, whichever occurred first. At baseline, imaging of the chest, abdomen, pelvis, and brain using computed tomography or magnetic resonance imaging was mandatory. Tumor assessments based on imaging of the chest, abdomen, and pelvis of all patients, as well as brain imaging for patients with brain metastasis at baseline, were conducted every 6 weeks during the first 54 weeks after randomization and every 12 weeks thereafter. For patients who had no brain metastasis, subsequent brain imaging was conducted at the discretion of investigators based on central nervous system symptoms of patients. Tumor response was assessed according to RECIST v1.1 by the independent radiology review committee (IRRC) and the investigators.

Adverse events and laboratory abnormalities were graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events, version 5.0. Adverse events were continuously monitored and collected for a duration of 30 days and serious adverse events (SAEs) and immune-related AEs were collected for 90 days after the last dose or the initiation of new antitumor treatment, whichever occurred first. Demographic variables, including age and sex, were collected at baseline.

Outcomes

The primary end point was IRRC-assessed PFS, defined as the time from the date of randomization to disease progression according to RECIST version 1.1 assessed by the IRRC or death from any cause (whichever occurred first) in the intention-to-treat (ITT) population. The secondary end points included overall survival; IRRC-assessed ORR, duration of response, disease control rate, and time to response; investigator-assessed PFS, ORR, duration of response, disease control rate, and time to response; safety; pharmacokinetic profile; and immunogenicity assessment. The results of pharmacokinetic profile and immunogenicity assessment will be reported in a separate publication.

Statistical Analysis

Approximately 320 patients were required for a total of 225 PFS events to provide 89% power to detect an HR of 0.65 favoring ivonescimab plus chemotherapy compared with placebo plus chemotherapy, with a 2-sided α of .05. An interim analysis of PFS was planned when 158 (70%) PFS events were observed. The Lan-DeMets spending function was used to estimate the O'Brien-Fleming boundary to control the overall 2-sided type I error at .05. Efficacy analyses of PFS included all randomized patients according to the ITT principle (Supplement 2).

Patients whose disease had not progressed or who died were censored at their last evaluable tumor assessment date. The median PFS time and the corresponding 95% CIs were estimated using the Kaplan-Meier method. A stratified log-rank test was used to compare the PFS between treatment groups. HRs and 95% CIs were calculated based on a stratified Cox proportional hazards model. The stratification factors were consistent with those used for randomization. The prespecified subgroup analyses of PFS were conducted for the baseline factors. The HRs and corresponding 95% CIs of the subgroups were estimated from an unstratified Cox model. The time-to-event variables of secondary end points were estimated using the Kaplan-Meier method. The Clopper-Pearson method was used to calculate the 95% CIs for the objective response rate and disease control rate for each treatment group. Safety data were summarized descriptively in the safety analysis set.

This interim analysis was triggered after 179 IRRC-assessed PFS events were observed. The updated O'Brien-Fleming αboundary at this interim analysis was .024. All statistical analyses were done using SAS, version 9.4 (SAS Institute). An independent data and safety monitoring committee reviewed efficacy and safety data for the prespecified interim analysis.

Results

Patients

In total, 501 patients were screened for enrollment at 55 sites in China. Between January 2022 and November 2022, a total of 322 patients were enrolled and randomized to receive ivonescimab plus chemotherapy or placebo plus chemotherapy. At data cutoff (March 10, 2023), 84 patients in the ivonescimab plus chemotherapy group and 62 patients in the placebo plus chemotherapy group continued receiving treatment, while 77 in the ivonescimab group vs 99 in the placebo group discontinued treatment, mainly due to disease progression (**Figure 1**). Baseline characteristics were well balanced between the 2 groups; the median (range) age was 59.6 (32.3-74.9) years in the ivonescimab group vs 59.4 (36.2-74.2) years in the placebo group. Additionally, in the ivonescimab group vs the placebo group, 85.1% vs 78.9% of patients had an Eastern

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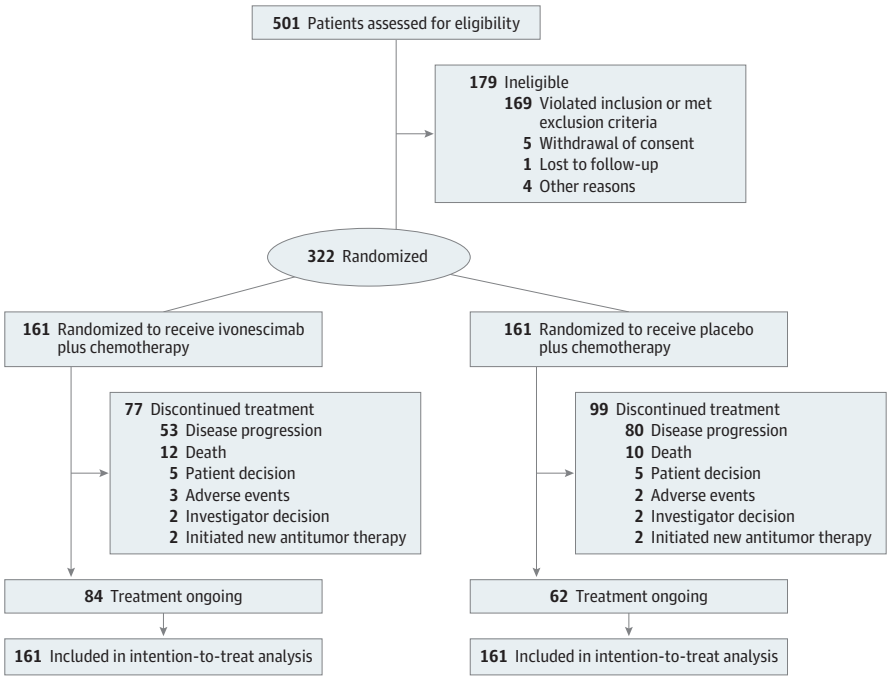
Figure 1. Flow of Patients in a Study of Ivonescimab Plus Chemotherapy in Non-Small Cell Lung Cancer



Cooperative Oncology Group performance score of 1 and 35 (21.7%) vs 37 (23.0%) had brain metastases. A total of 139 patients (86.3%) in the ivonescimab group vs 137 (85.1%) in the placebo group received third-generation EGFR-TKI treatment (**Table 1**).

Efficacy

The median duration of follow-up was 7.9 months. With 179 events of disease progression or death, the median PFS with ivonescimab plus chemotherapy was 7.1 (95% CI, 5.9-8.7) months vs 4.8 (95% CI, 4.2-5.6) months with placebo plus chemotherapy (difference, 2.3 months; HR, 0.46 [95% CI, 0.34-0.62]; $P < .001$) assessed by IRRC (**Figure 2**). In the ivonescimab group vs the placebo group, the 3-month PFS rate was 94.7% (95% CI, 89.8%-97.3%) vs 69.7% (95% CI, 61.7%-76.3%), the 6-month PFS rate was 55.4% (95% CI, 46.1%-63.7%) vs 33.1% (95% CI, 25.0%-41.3%), and the 9-month PFS rate was 37.9% (95% CI, 27.6%-48.2%) vs 18.3% (95% CI, 11.0%-27.1%).

The overall survival data were not complete, but at the time of this PFS analysis, 69 patients (21.4%) died: 32 (19.9%) in the ivonescimab group and 37 (23%) in placebo group. In ITT population, the ORR assessed by IRRC was 50.6% (95% CI, 42.6%-58.6%) with invonescimab vs 35.4% (95% CI, 28.0%-43.3%) with placebo (difference, 15.6% [95% CI, 5.3%-26.0%]; $P = .006$). The median duration of response was 6.6 (95% CI, 4.3-7.6) months in the ivonescimab group vs 4.2 (95% CI, 3.0-4.7) months in the placebo group; the 3-month duration of response rate was 79.3% (95% CI, 67.4%-87.2%) vs 64.8% (95% CI, 49.8%-76.3%) and the 6-month duration of response rate was 51.5% (95% CI, 37.4%-63.9%) vs 24.4% (95% CI, 11.3%-40.2%). Disease control was achieved in 149 of 160 patients (93.1% [95% CI, 88%-96.5%]) in the ivonescimab group vs 134 of 161 (83.2% [95% CI, 76.5%-88.6%]) in the placebo group. The median time to response was 1.4 (95% CI, 0.8-6.9) months in the ivonescimab group vs 1.4 (95% CI, 1.2-7.2) months in the placebo group (eTable 1 in Supplement 3).

The median PFS as assessed by investigators was consistent with the results assessed by IRRC; the median PFS was 7.0 (95% CI, 5.9-7.8) months in the ivonescimab group vs 4.8 months (95% CI, 4.3-5.6) in the placebo group (difference, 2.2 months; HR, 0.53 [95% CI, 0.40-0.71]; $P < .001$) (eFigure 1 in Supplement 3). Investigator-assessed treatment responses were consistent with IRRC-assessed results.

A subgroup analysis showed PFS benefit favoring patients with ivonescimab over placebo across almost all prespecified subgroups (**Figure 3**). In patients whose disease progressed while receiving third-generation EGFR-TKI therapy, ivonescimab plus chemotherapy significantly improved PFS over chemotherapy alone (HR, 0.48 [95% CI, 0.35-0.66]). Patients with *EGFR* variant of deletion 19 had an HR of 0.48 (95% CI, 0.32-0.73) and patients with T790M variant had an HR of 0.22 (95% CI, 0.09-0.54). For patients who had brain metastasis at baseline, adding ivonescimab to chemotherapy led to greater PFS benefit than chemotherapy alone (HR, 0.40 [95% CI, 0.22-0.73]). The median PFS was 5.75 (95% CI, 5.45-8.74) vs 4.14 (95% CI, 2.73-5.42) months in patients with brain metastasis and 7.36 (95% CI, 5.98 to not estimable) vs 5.32 (95% CI, 4.30-5.72) months in patients without metastasis. The Kaplan-Meier plots for IRRC-assessed PFS in these patients were shown in Figure 2B and Figure 2C.

Safety

All patients received at least 1 dose of the study treatment. The median duration of treatment exposure was 8.8 cycles for

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ivonescimab and 7.3 cycles for placebo. Details of treatment exposure are summarized in eTable 2 in Supplement 3.

Treatment-emergent AEs (TEAEs) occurred in 160 patients (99.4%) in the ivonescimab group and 157 patients (97.5%) in the placebo group. Grade 3 or higher TEAEs occurred in 99 patients (61.5%) in the ivonescimab group vs 79 patients (49.1%) in the placebo group. The most common grade 3 or higher TEAEs were neutrophil count decreased (31.1% vs 19.3%), white blood cell count decreased (20.5% vs 16.8%), platelet count decreased (16.1% vs 12.4%), and anemia (15.5% vs 14.3). The TEAEs with an incidence of greater than or equal to 10% are shown in **Table 2**. Serious AEs (SAEs) occurred in 66 patients (41.0%) in the ivonescimab group vs 41 (25.2%) in the placebo group. The most common SAEs were disease progression (10.6% vs 6.8%), platelet count decreased (7.5% vs 4.3%), anemia (3.7% vs 2.5%), infectious pneumonia (3.7% vs 0.6%), COVID-19 infection (3.1% vs 1.2%), and abnormal liver function (2.5% vs 0%). The summary of adverse events is shown in eTable 3 in Supplement 3. The details of SAEs were listed in eTable 4 in Supplement 3.

TEAEs leading to discontinuation of ivonescimab occurred in 9 patients (5.6%) vs discontinuation of placebo in 4 patients (2.5%). The 9 TEAEs leading to ivonescimab discontinuation included elevated alanine aminotransferase, elevated serum creatinine, musculoskeletal pain, hypothyroidism, rash, kidney dysfunction, heart failure, arterial embolism, and glaucoma. TEAEs leading to death occurred in 17 patients (10.6%) receiving ivonescimab, and the causes of death were all disease progression. There were 12 (7.5%) TEAEs leading to death in the placebo group, and the causes of 11 deaths were disease progression and 1 death was thermal burn.

Immune-related AEs occurred in 39 patients (24.2%) in the ivonescimab group vs 10 patients (6.2%) in the placebo group; grade 3 or higher immune-related AEs occurred in 6.2% vs 2.5% of patients. The most common grade 3 or higher immune-related AEs were rash (2.5% vs 0.6%), dermatitis (1.2% vs 0%), interstitial lung disease (1.2% vs 0.6%), lung inflammation (0.6% vs 0%), liver dysfunction (0.6% vs 0%), and hypothyroidism (0.6% vs 0%) (eTable 5 in Supplement 3).

The AEs of special interest (AESI) related to VEGF blocking were also evaluated. A total of 48 patients (29.8%) reported an AESI in the ivonescimab group vs 25 (15.5%) in the placebo group. Most of the AESIs were grade 1 to 2. Grade 3 or higher AESIs occurred in 5 patients (3.1%) in the ivonescimab group vs 4 patients (2.5%) in the placebo group. The most common AESIs were proteinuria (17.4% vs 8.1%), bleeding (6.8% vs 5.0%), and hypertension (8.1% vs 3.1%). All bleeding events were grade 1 to 2. Grade 3 or higher proteinuria occurred in 1 patient and grade 3 or higher hypertension occurred in 3 patients in each group (eTable 6 in Supplement 3).

Table 1. Baseline Patient Characteristics

	Ivonescimab plus chemotherapy (n = 161)	Placebo plus chemotherapy (n = 161)
Age, median (range), y	59.6 (32.3-74.9)	59.4 (36.2-74.2)
Age ≥65 y, No. (%)	50 (31.1)	51 (31.7)
Sex, No. (%)		
Female	84 (52.2)	82 (50.9)
Male	77 (47.8)	79 (49.1)
ECOG, No. (%)[a]		
0 (Fully active with no performance restriction)	24 (14.9)	34 (21.1)
1 (Ambulatory but strenuous physical activity restricted)	137 (85.1)	127 (78.9)
Disease stage, No. (%)[b]		
IIIB/C	3 (1.9)	5 (3.1)
IV	158 (98.1)	156 (96.9)
Brain metastases, No. (%)	35 (21.7)	37 (23.0)
Liver metastases, No. (%)	21 (13.0)	17 (10.6)
Distant metastases ≥3, No. (%)	74 (46.0)	68 (42.2)
EGFR variant, No. (%)		
Exon 19 deletion	92 (57.1)	78 (48.4)
L858R	60 (37.3)	78 (48.4)
Other	35 (21.7)	25 (15.5)
T790M variant, No. (%)		
Negative	26 (16.1)	27 (16.8)
Positive	26 (16.1)	18 (11.2)
Unknown	109 (67.7)	116 (72.0)
Generation of previous EGFR-TKI therapy, No. (%)		
First-line first-/second-generation TKI	22 (13.7)	24 (14.9)
First-line third-generation TKI	49 (30.4)	58 (36.0)
First-line first-/second-generation TKI followed by second-line third-generation TKI	90 (55.9)	79 (49.1)
Third-generation TKIs, No. (%)		
Osimertinib	97 (60.2)	92 (57.1)
Almonertinib	43 (26.7)	44 (27.3)
Furmonertinib	12 (7.5)	8 (5)

Abbreviation: EGFR-TKI, epidermal growth factor receptor tyrosine kinase inhibitor.

[a] Eastern Cooperative Oncology Group (ECOG) performance status scores range from 0 to 5 (higher scores indicate greater disability).

[b] Stage at diagnosis was based on *American Joint Committee on Cancer Staging Manual, Eighth Edition*.

Discussion

To the authors' knowledge, this is the first phase 3 study to show a notable clinical benefit of ivonescimab plus chemotherapy in patients whose disease progressed while receiving EGFR-TKIs treatment. This study demonstrated that ivonescimab plus chemotherapy led to a significant longer PFS duration than chemotherapy alone (HR, 0.46; $P < .001$). The rate of PFS was higher with ivonescimab plus chemotherapy than chemotherapy alone at 3 months, 6 months, and 9 months.

Third-generation TKIs have been the standard of care for patients with NSCLC that harbor the *EGFR* variant in most regions.[3,17] However, patients inevitably develop resistance. Currently the standard of care for patients who experience resistance to third-generation TKIs is platinum-based double chemotherapy as per European Society for Medical Oncology and

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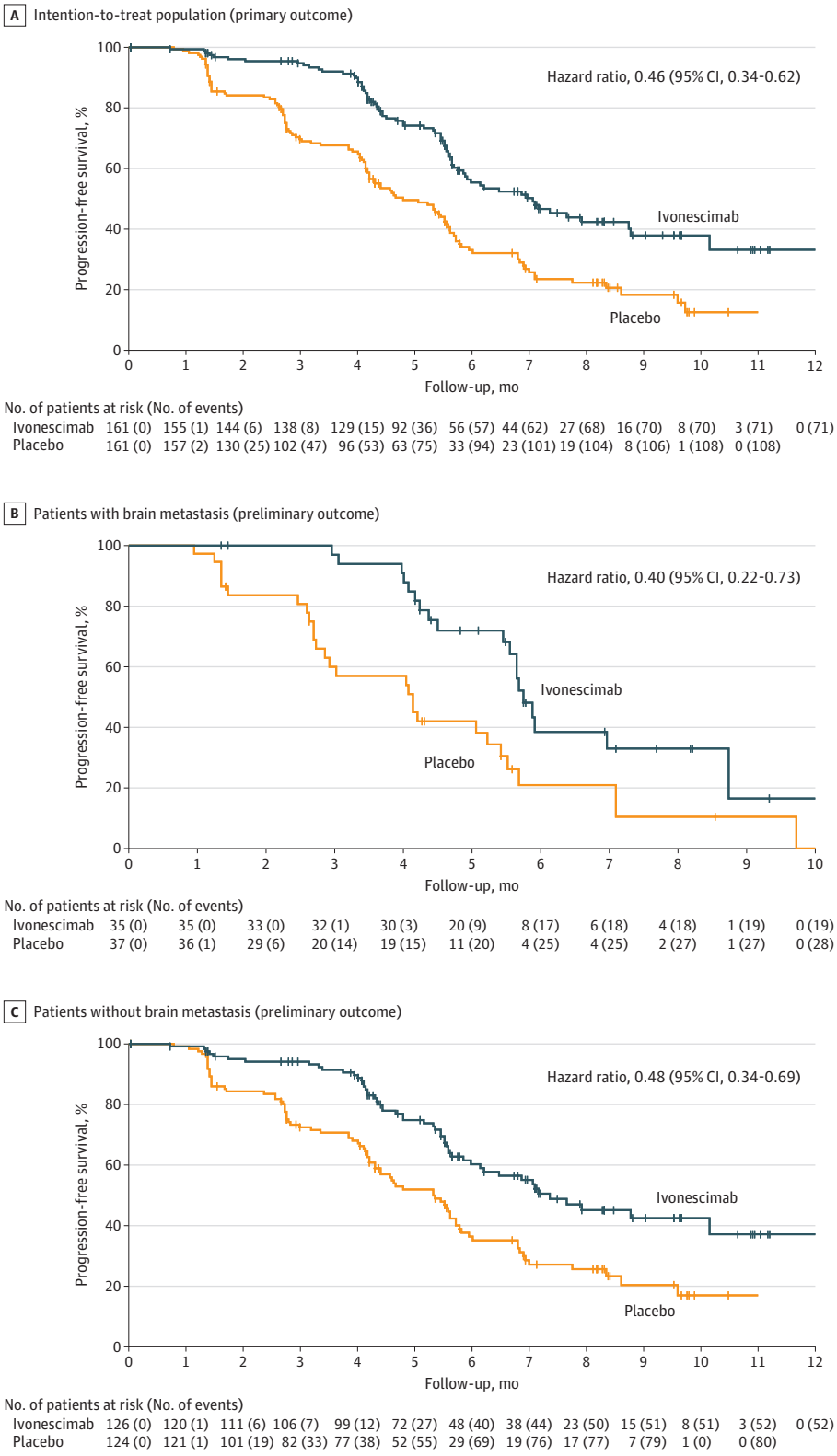
Figure 2. Independent Radiographic Review Committee–Assessed Progression-Free Survival



A, Median (IQR) follow-up time: 7.1 (5.4-9.0) months for ivonescimab and 8.2 (5.5-9.5) months for placebo. B, Median (IQR) follow-up time: 7.1 (5.5-8.2) months for ivonescimab and 8.5 (4.3 to not estimable) months for placebo. C, Median (IQR) follow-up time: 7.1 (5.4-9.0) months for ivonescimab and 8.2 (5.5-9.5) months for placebo.

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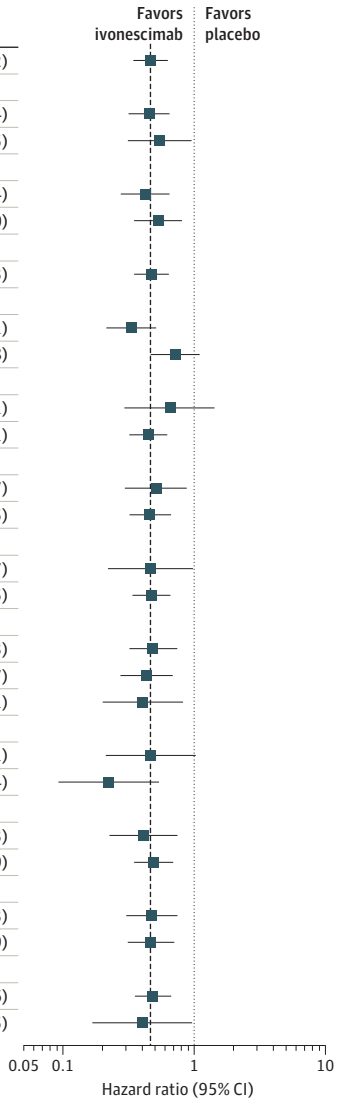
Figure 3. Subgroup Analysis of Progression-Free Survival

Characteristic	No. of events/No. of participants: Ivonescimab	No. of events/No. of participants: Placebo	Hazard ratio (95% CI)
All patients	71/161	108/161	0.46 (0.34-0.62)
Age, y			
<65	51/111	75/110	0.45 (0.31-0.64)
≥65	20/50	33/51	0.54 (0.31-0.95)
Sex			
Male	34/77	57/79	0.41 (0.27-0.64)
Female	37/84	51/82	0.52 (0.34-0.80)
Clinical stage at study entry			
IV	69/158	105/156	0.47 (0.34-0.63)
No. of distant metastasis sites at baseline			
<3	30/87	64/93	0.33 (0.21-0.51)
≥2	41/74	44/68	0.70 (0.46-1.08)
Liver metastasis			
Presence	13/21	12/17	0.64 (0.29-1.41)
Absence	58/140	96/144	0.44 (0.32-0.61)
Smoking history			
Yes	23/49	31/46	0.50 (0.29-0.87)
No	48/112	77/115	0.45 (0.32-0.65)
Baseline ECOG score			
0	10/24	22/34	0.46 (0.22-0.97)
1	61/137	86/127	0.47 (0.33-0.65)
Baseline *EGFR* variation			
Exon 19 deletion	39/92	53/78	0.48 (0.32-0.73)
L858R	29/60	54/78	0.43 (0.27-0.67)
Other	15/35	17/25	0.40 (0.20-0.81)
T790M variation status			
Negative	10/26	17/27	0.46 (0.21-1.01)
Positive	12/26	13/18	0.22 (0.09-0.54)
Baseline brain metastasis			
Presence	19/35	28/37	0.40 (0.22-0.73)
Absence	52/126	80/124	0.48 (0.34-0.69)
Previously received EGFR-TKI treatment			
First-line	30/71	52/82	0.47 (0.30-0.73)
Second-line	41/90	56/79	0.46 (0.31-0.69)
Previous treatment with third-generation EGFR-TKI			
Recived	63/139	94/137	0.48 (0.35-0.66)
Not received	8/22	14/24	0.40 (0.17-0.95)



Chemotherapy consisted of pemetrexed and carboplatin. Progression-free survival was evaluated by an independent radiographic review committee. The dashed line notes the hazard ratio of 0.46 (overall progression-free survival). ECOG indicates Eastern Cooperative Oncology Group; EGFR-TKI, epidermal growth factor receptor tyrosine kinase inhibitor.

National Comprehensive Cancer Network guideline. In the current study in which 86% of patients received third-generation TKI treatment, ivonescimab plus chemotherapy significantly improved PFS in the ITT population. The subgroup analysis showed consistent results that ivonescimab plus chemotherapy significantly improved PFS in patients who had received third-generation TKI treatment, regardless of whether the third-generation TKI was used as first- or second-line treatment.

The benefit of ivonescimab in terms of PFS was consistent across all prespecified subgroups. Significant PFS benefit was observed in patients with exon 19 deletion (HR 0.52 [95% CI, 0.35-0.78]), the L858R point variant (HR, 0.51 [95% CI, 0.33-0.80]), and the T790M variant (HR, 0.24 [95% CI, 0.09-0.59]). In the subgroup analysis in the ATTLAS study,[11] perhaps because of the small sample size, significant benefit was not observed in patients harboring exon 19 deletion (HR, 0.69 [95% CI, 0.44-1.08]) or the T790M variant (HR, 1.07 [95% CI, 0.59-1.94]). Similarly, in the subgroup analysis in the ORIENT 31 trial, significant benefit with sintilimab plus IBI305 and chemotherapy was not observed in patients with the T790M variant (HR, 0.84 [95% CI, 0.50-1.43])[12] and, in the sintilimab plus chemotherapy group, no benefit was observed in patients harboring exon 19 deletion (HR, 1.04 [95% CI, 0.73-1.47]) or the T790M variant (HR, 1.10 [95% CI, 0.68-1.79]).[18] Furthermore, preclinical study observed that in the presence of VEGF, the binding avidity of ivonescimab to PD-1 was able to increase 10-fold.[14] Although the results from this subgroup analysis should be discreetly interpreted, different results from the above studies may suggest that ivonescimab has different mechanisms of action compared with the combination of monoclonal antibody, and the strength of synergistic effect between the 2 targets may influence the clinical efficacy.

The PFS benefit was also observed in patients with brain metastases, who account for 46% to 57% of patients with advanced NSCLC,[19-21] which was consistent with the results of subgroup analyses from ORIENT-31 and ATTLAS studies.

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Table 2. Treatment-Emergent Adverse Events[a]

	Ivonescimab plus chemotherapy, No. (%)		Placebo plus chemotherapy, No. (%)	
	Any grade	Grade ≥3	Any grade	Grade ≥3
White blood cell count decreased	106 (65.8)	33 (20.5)	104 (64.6)	27 (16.8)
Anemia	102 (63.4)	25 (15.5)	118 (73.3)	23 (14.3)
Neutrophil count decreased	97 (60.2)	50 (31.3)	94 (58.4)	31 (19.3)
Platelet count decreased	78 (48.4)	26 (16.1)	66 (41.0)	20 (12.4)
Aspartate aminotransferase increased	71 (44.1)	2 (1.2)	50 (31.1)	3 (1.9%)
Alanine aminotransferase increased	60 (37.3)	4 (2.5)	59 (36.6)	4 (2.5)
Vomiting	55 (34.2)	2 (1.2)	57 (35.4)	0
Decreased appetite	53 (32.9)	0	35 (21.7)	0
COVID-19 infection	49 (30.4)	3 (1.9)	33 (20.5)	1 (0.6)
Constipation	46 (28.6)	0	32 (19.9)	0
Fatigue	36 (22.4)	3 (1.9)	28 (17.4)	2 (1.2)
γ-Glutamyl transferase increased	35 (21.7)	5 (3.1)	22 (13.7)	1 (0.6)
Proteinuria	35 (21.7)	1 (0.6)	21 (13.0)	0
Hypoalbuminemia	33 (20.5)	0	23 (14.3)	0
Hypercholesterolemia	33 (20.5)	0	20 (12.4)	0
Blood glucose increased	31 (19.3)	2 (1.2)	19 (11.8)	0
Hypertriglyceridemia	29 (18.0)	0	20 (12.4)	0
Weight decrease	16 (16.1)	2 (1.2)	17 (10.6)	0
Blood lactate dehydrogenase increased	24 (14.9)	0	25 (15.5)	0
Lymphocyte count decreased	23 (14.3)	13 (8.1)	16 (9.9)	6 (3.7)
Hypothyroidism	23 (14.3)	0	2 (1.2)	0
Hypokalemia	19 (11.8)	3 (1.9)	17 (10.6)	3 (1.9)
Blood alkaline phosphatase increased	19 (11.8)	0	17 (10.6)	0
Hyperlipemia	18 (11.2)	0	7 (4.3)	0
Fever	19 (11.8)	0	14 (8.7)	0
Amylase increased	17 (10.6)	2 (1.2)	18 (11.2)	0
Rash	17 (10.6)	5 (3.1)	8 (5.0)	1 (0.6)

[a] All treatment-emergent adverse events were determined according to Common Terminology Criteria for Adverse Events, version 5.0. Grade 3 is defined as delay in symptom relief (eg, inability to respond quickly to symptomatic treatment and/or infusion interruption), recurrence after improvement of symptoms, hospitalization required for sequelae; grade 4, life-threatening, urgent treatment indicated; and grade 5, death.

Bevacizumab could delay or prevent progression of brain metastases in NSCLC.[22,23] Improved PFS in the patients with brain metastases in the current study may attribute to the blocking of VEGF and/or a combined effect of anti-PD-1/VEGF of the bispecific antibody ivonescimab.

MARIPOSA-2 was a phase 3 study designed to evaluate amivantamab plus chemotherapy with and without lazertinib in *EGFR*-variant advanced NSCLC after disease progression while receiving osimertinib. The results showed that PFS was significantly longer for amivantamab plus chemotherapy and amivantamab plus lazertinib and chemotherapy vs chemotherapy (median of 6.3 and 8.3 vs 4.2 months) and grade 3 or higher AEs occurred in 72% of patients treated with amivantamab plus chemotherapy and 92% of patients with amivantamab plus lazertinib and chemotherapy.[24] In the current study, ivonescimab plus carboplatin and pemetrexed were well tolerated with no new safety signals. The incidence of grade 3 or higher TEAEs with ivonescimab (61.5%) was higher than with chemotherapy alone (49.1%); most of the AEs were related to cytotoxic chemotherapy and it seems that chemotherapy-related grade 3 or higher TEAEs were more frequent in the ivonescimab group, such as white blood cell decrease (20.5% vs 16.8%), neutrophil count decrease (31.3% vs 19.3%), and platelet count decrease (16.1% vs 12.4%). Nonetheless, the TEAEs were manageable with appropriated supportive care. The incidences of grade 3 or higher TEAEs related to anti-PD-1/VEGF were similar between treatment and placebo groups. The grade ≥3 TEAEs in the ivonescimab group in the current study was comparable with that in the ORIENT-31 study (55%) and the IMpower 151 study (66%), in which atezolizumab plus bevacizumab, carboplatin, and pemetrexed were used.[13] It was worth noting that the incidences of grade 3 or higher immune-related AEs and grade 3 or higher AESIs were similar in the ivonescimab and placebo group. In comparison with other studies in which a 4-drug combination regimen was utilized, this study reported lower immune-related AEs, with an incidence of 24.2%. This contrasts with the ORIENT-31 study, which reported an incidence rate of 41%, and the ATLLAS study, in which the combination of atezolizumab plus bevacizumab and chemotherapy resulted in an incidence of 54.2%. In particular, the incidence of hypertension in the current study was much lower (n = 13 [8.1%]) compared with that in the ORIENT-31 study (n = 36 [24%]) and the ATLLAS study (n = 30 [19.9%]). Most of the AESIs that were related to VEGF blocking were grade 1 to 2. All of the results suggest that in terms of immune-related and antivascularity-related safety, ivonescimab plus chemotherapy has an advantage over anti-PD-1 plus bevacizumab and chemotherapy.

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Limitations

There are some limitations in this study. First, the overall survival data were not complete at data cutoff. Second, the preplanned subgroup analysis showed promising efficacy of ivonescimab plus chemotherapy in specific populations; however, the sample size in the subgroups were relatively small, so the current findings should be interpreted with caution. Third, biomarker analyses, including the correlation between PD-L1 expression and efficacy, requires further exploration.

Conclusions

Ivonescimab plus chemotherapy significantly improved PFS in patients with NSCLC whose disease progressed while receiving EGFR-TKI treatment. The safety profile was tolerable and manageable. Several phase 3 trials with ivonescimab monotherapy and combination therapy in NSCLC are ongoing.

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